

THREE PARADIGMS OF PHYSICAL WORLD MODELING

Contrasting Generative Pixel Diffusion, Latent State Transitions (JEPA/RSSM), and Deterministic SE(3) Coordinate Trees

1. GENERATIVE PIXEL DIFFUSION

Video Diffusion (Sora / Genie)

x_t (Pixels) + $a_t \rightarrow 50$ Denoising Steps

Output: $\hat{x}_{t+1}$ (Rendered Video)

Computes all background pixels & lighting

Latency (P50/P99): 450 / 1200 ms ❌

Memory Footprint: 16+ GB VRAM

Compute Load: > 30 TFLOPs

FATAL EMBEDDED DEFECTS:

- **Pixel Hallucinations:** Objects phase through solid walls or change material properties mid-frame
- **Freshness Wall Violation:** Latency exceeds 60 ms by 20x, guaranteeing stale world state
- **Compute Starvation:** Consumes 100% of SoC power, causing thermal throttling

VERDICT FOR PHYSICAL AI:

❌ Inadmissible for Real-Time Control

2. LATENT DYNAMICS (JEPA / RSSM)

V-JEPA / DreamerV3 RSSM

$E_\theta(x_t) = z_t \in Z$ (Feature Space)

Transition: $\hat{z}_{t+1} = f_\phi(z_t, a_t)$

Predicts semantics without pixel rendering

Latency (P50/P99): 2.2 / 3.8 ms ✓

Memory Footprint: 420 MB DRAM

Compute Load: 0.08 TFLOPs

SYSTEMS ADVANTAGES:

- **150x Faster:** Skips high-overhead pixel decoders; executes direct latent transitions
- **Noise Invariance:** Ignores dynamic background clutter, shadows, and occlusions
- **Action Evaluation:** Rolls out 100 candidate trajectories in parallel

VERDICT FOR PHYSICAL AI:

✓ Ideal for System 1.5 MPC Planning

3. METRIC SE(3) TREE + LIE ALGEBRA

Dynamic Transform Tree (SE(3))

$T_A^B = [R | p] \in SE(3)$, Covariance Σ

Tangent: $T(t) = T(t_0) \cdot \exp(\hat{\xi} \Delta t)$

Zero singularities; exact metric coordinates

Latency (P50/P99): < 0.05 ms (50 μ s) *

Memory Footprint: < 64 KB SRAM

Determinism: Zero Dynamic Allocations

SYSTEMS ADVANTAGES:

- **100% Deterministic:** Runs inside 1 kHz FreeRTOS reflex loop on microcontroller
- **Mathematical Safety Barrier:** Computes exact metric distance d_{min} to obstacles
- **Sensory Lag Compensation:** Time-indexed buffer rewinds and forecasts

VERDICT FOR PHYSICAL AI:

✓ Mandatory for System 1 Safety Reflex